Balancing Accuracy and Efficiency for Road Intersection Detection Method in Large-Scale High-Resolution Remote Sensing Imagery

Yuhao Tang, Bo Liu, Youneng Su, Hua Liu, Yuanyuan Liu, and Yuanwei Yang

***Abstract*—Road intersections serve as critical topological nodes in transportation networks, and their rapid, accurate identification is essential for intelligent transportation systems and geographic information updates. However, direct detection of road intersections on large-scale high-resolution remote sensing imagery remains challenging due to intersection structural diversity, complex illumination conditions, and vegetation occlusion, which make it difficult to balance detection accuracy and computational efficiency. To address these challenges, Swin Transformer-GSConv-efficient IoU-YOLO (SW-GSE-YOLO), a novel lightweight and high-precision road intersection detection model, is proposed. The model incorporates three key innovations: 1) a Swin-C2PSA module that is designed by integrating the shifted-window self-attention mechanism of the Swin Transformer with the C2PSA structure, enabling efficient large-scale feature modeling while enhancing local detail and global contextual representation; 2) a GS3k2 lightweight network architecture is built upon GSConv and CSPNet, which reduces the parameter count while preserving feature extraction capability; and 3) the efficient intersection over union (EIoU) loss function is adopted to optimize bounding box regression and improve intersection center localization accuracy. To validate the proposed method, the RIS-42k dataset is constructed, comprising 42 372 annotated image patches of size 400 $\times$ 400 pixels. Experimental results demonstrate that SW-GSE-YOLO outperforms recent mainstream lightweight detectors, including You Only Look Once (YOLO)v8–YOLOv12 and real-time DEtection TRansformer (RT-DETR). Compared with the YOLO11n baseline, it achieves improvements of 5.9%, 1.3%, 3.9%, and 14.6% in precision, recall, mAP at 0.5, and mAP at [0.5:0.95], respectively, while significantly reducing GPU memory usage and inference latency for large-scale image processing. Furthermore, SW-GSE-YOLO demonstrates robust localization accuracy and strong generalization capability across complex urban, suburban, and rural environments.**

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***Index Terms*—Efficient intersection over union (EIoU), GSConv, high-resolution remote sensing imagery, road intersection, Swin Transformer-GSConv-efficient IoU-YOLO (SW-GSE-YOLO), Swin-C2PSA.**

I. Introduction

ROAD intersections are critical hub nodes in urban road networks and core areas for the convergence and dispersal of vehicle and pedestrian flow. The rapid and accurate identification of these nodes is essential for applications such as map updating, intelligent navigation, and traffic management. While high-resolution remote sensing imagery offers rich geometric and textural details that facilitate intersection localization, it also exacerbates challenges related to occlusions, shadows, and complex backgrounds. These interference factors often result in fragmented road features and spectral inconsistency. Consequently, a key challenge in current research is how to effectively leverage the spatial fidelity of high-resolution imagery while mitigating the adverse effects of environmental noise.

Furthermore, road intersections in real-world scenarios exhibit substantial morphological diversity (such as T-shaped, cross-shaped, Y-shaped intersections, and roundabouts), and their sizes vary significantly across different regions (e.g., urban and suburban) [1]. This diversity further increases the difficulty of identifying and extracting road intersections from high-resolution remote sensing imagery. Traditional road intersection extraction methods primarily rely on morphological operations, template matching, or rule-based feature extraction [2], [3], [4]. These methods often require the design of complex feature operators and decision rules tailored to specific scenarios, resulting in limited generalization ability and difficulty in adapting to the diverse variations in intersection morphology and scale. In addition, their levels of automation and processing efficiency are generally low.

In recent years, with the rapid development of deep learning technology, object detection methods represented by the R-CNN series [5], [6], [7], SSD [8], the You Only Look Once (YOLO) series [9], [10], [11], [12], [13], and DETR-based frameworks [14] have achieved great success in natural image understanding. Owing to their strong learning and representation capabilities, these methods have also been increasingly introduced into intelligent interpretation tasks for remote sensing imagery [15]. Among them, the YOLO

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series algorithms have attracted widespread attention because it offers a favorable balance between detection accuracy and inference speed.

However, high-resolution remote sensing imagery is characterized by an overhead imaging perspective and complex spatial contextual relationships. As a result, directly applying generic YOLO models to road intersection recognition still faces substantial challenges, which are mainly reflected in the following aspects.

1) *Contradiction Between Detection Accuracy and Computational Efficiency:* Although CNN-based detectors such as YOLOv5 and YOLOv8 can process large-scale remote sensing imagery, their feature modeling capability is constrained by the limited receptive field of convolution operations, making it difficult to effectively capture the long-range spatial dependencies that road intersection recognition often relies on. To alleviate this issue, YOLO11 introduces the C2PSA module, which expands the effective receptive field and enhances recognition capability via attention mechanisms. However, C2PSA adopts global self-attention, whose computational complexity scales quadratically with the feature map area, i.e., $O((HW)^2)$. When directly processing large, high-resolution remote sensing imagery, the computational cost increases dramatically. Using image tiling to reduce computation inevitably disrupts the global structural continuity of the road network, thereby making it difficult to simultaneously achieve high detection accuracy and high computational efficiency in large-scale scenarios.
2) *Insufficient Center Point Localization Accuracy:* Generic YOLO models localize objects primarily through bounding box regression. In contrast, road intersections in remote sensing imagery inherently correspond to topological nodes with a well-defined center, and their spatial positions often require high accuracy in practical applications. Relying solely on bounding box regression, therefore, makes it difficult to achieve precise localization of road intersection centers, which limits the effectiveness of such models in real-world deployment.

To address the aforementioned challenges, this article proposes Swin Transformer-GSConv-efficient IoU-YOLO (SW-GSE-YOLO), a road intersection detection model for large-scale high-resolution remote sensing imagery that balances detection accuracy and computational efficiency, built upon the YOLO11 framework. The main contributions of this article are summarized as follows.

1) A Swin-C2PSA module tailored for large-scale high-resolution remote sensing imagery is designed. By integrating the shifted-window attention mechanism of Swin Transformer with the C2PSA module in YOLO11n [16], [17], we propose a Swin-C2PSA module. This module confines self-attention computation within local windows, enabling the model to directly process large-scale high-resolution remote sensing imagery without image tiling. Meanwhile, it effectively enhances the model's joint perception of intersection-level local details and global contextual features, improving recognition accuracy while controlling computational cost.
2) An efficient lightweight network architecture is constructed: By introducing GSConv [18] and combining it with the CSP structure of YOLO11n [19], we design a GS3k2 module and integrate it into the network. This structure significantly reduces the parameter count and computational complexity while further enriching feature representation capability, thereby improving both efficiency and accuracy for road intersection detection in high-resolution remote sensing imagery.
3) Road intersection center point localization accuracy is improved: We introduce the efficient intersection over union (EIoU) loss function [20]. This loss jointly considers bounding box overlap, center-point distance, and width–height discrepancies, thereby improving the localization accuracy of road intersection centers in high-resolution remote sensing imagery.
4) Large-scale road intersection detection dataset (RIS-42k) is constructed: We construct a high-resolution remote sensing road intersection dataset (RIS-42k) containing 42 372 samples across diverse scenarios. Extensive comparative experiments and ablation studies conducted on this dataset comprehensively validate the proposed SW-GSE-YOLO model, demonstrating its advantages in detection accuracy and computational efficiency.

The remainder of this article is organized as follows. Section II reviews early methods for road intersection recognition, object detection frameworks, lightweight network design, attention mechanisms, and intersection over union (IoU)-related research. Section III details the network architecture and key modules of SW-GSE-YOLO. Section IV presents the dataset, experimental results, and analysis. Section V concludes this article and discusses future research directions.

II. Related Work

Currently, existing research methods for road intersection recognition and extraction can be broadly categorized into two groups: methods based on crowdsourced or open-source vector data and methods based on remote sensing imagery.

Methods based on crowdsourced or open-source vector data can be further divided into two categories. The first category directly utilizes the geometric shapes, topological relationships, and attribute information contained in vector data to recognize and extract road intersections. For example, Du et al. [21] designed point filtering rules for trajectory data and combined clustering and mean-based optimization methods to localize road intersection center points. Tang et al. [22] and Liu et al. [23] identified turning points or intersection-related points using rule-based approaches or classification models and further employed clustering algorithms to aggregate these points into intersection center locations. The second category first converts vector data into raster format and then applies image processing or deep learning techniques for road intersection recognition and extraction. For instance, Zhang et al. [24] achieved automatic road intersection detection by applying an improved YOLOv5 model to trajectory raster images. Li et al. [25] converted road vector data into raster imagery and employed the Faster R-CNN model to

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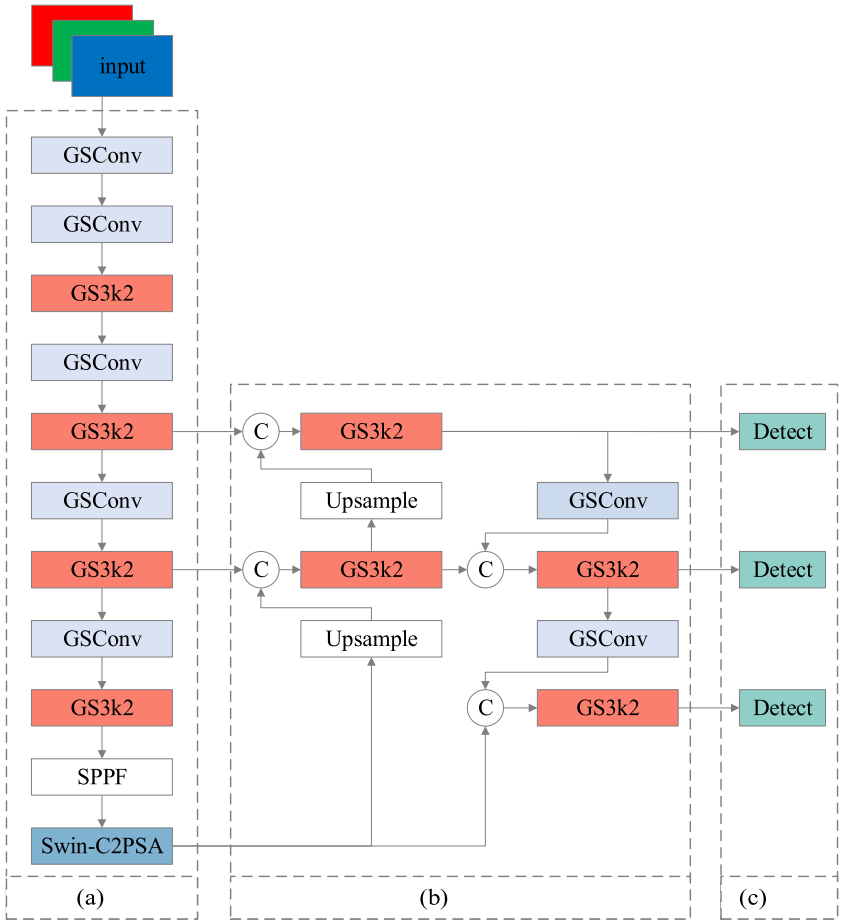


Fig. 1. Overview of SW-GSE-YOLO. (a) Backbone. (b) Neck. (c) Head.

identify overpass structures within road networks. Although this category of methods extends the application scope of deep learning models in road structure analysis to some extent, the quality of their results remains highly dependent on the completeness and accuracy of the original vector data.

In the early stage of development, road intersection recognition methods based on remote sensing imagery primarily relied on expert prior knowledge, using manually designed image features (e.g., gradients and textures) and geometric rules (e.g., circular search regions) for detection. Ahmed and Rahman [3] employed morphological filtering to separate road regions and subsequently extracted road intersections; Sun et al. [2] adopted template matching and set-based feature filtering to obtain initial road regions, followed by tensor voting to repair road structures and extract intersections; and Dai et al. [4] inferred road intersections by constructing predefined local search regions and fuzzy inference rules. However, the feature and rule design of these methods is highly dependent on specific scenarios. Once imaging conditions or regional environments change, their recognition performance often degrades significantly, resulting in limited generalization capability and practical applicability.

In recent years, with the rapid development of deep learning, models represented by convolutional neural networks (CNNs) have demonstrated the ability to automatically learn more robust features from large-scale training data, effectively overcoming the poor generalization limitations of traditional methods. As a result, these models have been widely applied in remote sensing image interpretation tasks. Among them, two-stage detection methods represented by Faster R-CNN achieve high localization accuracy through a processing framework consisting of region proposal generation followed by classification and bounding box regression [7]. Qin et al. [26] improved the detection accuracy of remote sensing targets by enhancing the feature extraction network of Faster R-CNN and incorporating data augmentation strategies. Wang et al. [27] applied Mask R-CNN to the road intersection detection task and obtained favorable detection performance. However, two-stage methods generally require candidate regions to be processed sequentially, leading to high computational overhead and slow inference speed, which makes them difficult to satisfy the efficiency requirements of large-scale remote sensing image processing in practical applications.

To overcome the speed bottleneck of two-stage detectors represented by Faster R-CNN, single-stage detectors represented by YOLO were proposed [9], which formulate object detection as an end-to-end regression problem. Liu et al. [28] proposed the MFAE-YOLO model, which improves the detection accuracy of multiscale targets in remote sensing imagery by introducing a global feature fusion module.

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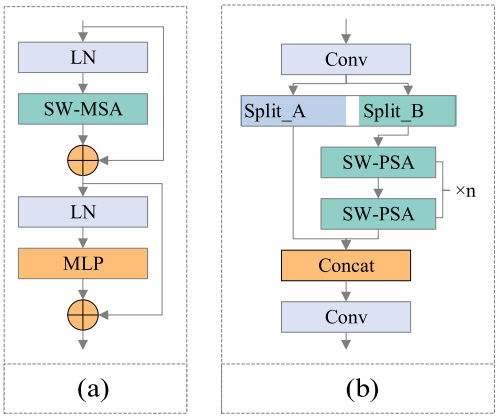


Fig. 2. Shift window attention module SW-PSA and Swin-C2PSA module. (a) SW-PSA. (b) Swin-C2PSA.

Shao et al. [29] and Xie et al. [30] enhanced the detection performance of road intersections and other ground objects in remote sensing imagery by incorporating improved loss functions and attention mechanisms within the YOLO framework. Yang et al. [31] proposed the IS-RoadDet model, achieving end-to-end vectorized road map detection. Eltaher et al. [32] achieved precise localization of road intersection center points based on the EfficientDet model.

Although detection accuracy for road intersections can be improved to some extent through network architecture optimization and data augmentation strategies, these models generally exhibit high network complexity and large parameter scales. As a result, they still face challenges in balancing detection accuracy and computational efficiency when applied to large-scale remote sensing imagery.

To address the issue of high computational overhead in deep learning-based remote sensing image processing, numerous studies have focused on model lightweighting and efficient feature modeling. MobileNet [33] significantly reduces computational cost by introducing depthwise convolution (DWConv). ShuffleNet [34] adopts group convolution and channel shuffle strategies, enabling model compression while preserving a certain level of feature representation capability. GhostNet [35] and EfficientNet [36] further achieve effective reductions in parameter count and computational complexity through cheap feature map transformations and compound scaling strategies, respectively.

Simultaneously, attention mechanism-based methods such as SENet [37] and CBAM [38] mitigate potential performance degradation introduced during the model lightweighting process by emphasizing discriminative features. Furthermore, self-attention mechanisms based on the Transformer architecture [39] have demonstrated strong potential for enhancing feature representation capability. Vision Transformer (ViT) [40] and Swin Transformer [16] have verified the effectiveness of Transformer-based architectures in various visual tasks. In particular, Swin Transformer [16], [41] significantly reduces computational complexity by introducing window-based self-attention and a hierarchical architecture, making it more suitable for processing large-scale remote sensing imagery. Wang et al. [42] employed Transformer-based models to extract global features from remote sensing imagery, thereby improving the structural completeness of road networks. Liu et al. [43] combined CNN and Transformer architectures to construct a hybrid network that fuses global and local information, further enhancing road extraction accuracy in remote sensing imagery.

In summary, although deep learning-based methods have achieved substantial progress in road intersection detection from remote sensing imagery, significant challenges remain when processing large-scale high-resolution data. Specifically, how to effectively control model computational complexity while ensuring detection accuracy, and simultaneously enhance model generalization capability, remains one of the key unresolved issues in current road intersection recognition research.

III. Method

A. Overview of SW-GSE-YOLO

Based on YOLO11n [17], the lightweight variant of YOLO11, this article proposes a novel lightweight detection model named SW-GSE-YOLO, whose overall network architecture is illustrated in Fig. 1. The proposed model effectively addresses the tradeoff between computational efficiency and detection accuracy encountered by YOLO11n when detecting road intersections in large-scale high-resolution remote sensing imagery.

First, to enhance the model's capability for processing large-scale remote sensing imagery, a Swin-C2PSA module is designed. By incorporating the shifted-window attention mechanism of the Swin Transformer, this module effectively expands the receptive field of the CNN components, thereby improving the large-scale scene modeling capability of the C2PSA module in YOLO11n.

Second, to further improve computational efficiency, GSConv [18] is introduced, based on which a GS3k2 module is designed to replace the standard Conv and C3k2 modules in both the backbone and neck networks of YOLO11n. While preserving feature extraction capability, this module significantly reduces the model's parameter count and computational complexity, thereby enhancing overall inference efficiency.

Finally, to improve the localization accuracy of road intersection center points, the EIoU loss function is adopted to replace the original complete intersection over union (CIoU) loss function in the YOLO11n detection head [20], [44]. This replacement accelerates model convergence and further enhances localization accuracy. The implementation details of these components are provided in Sections III-B–III-D.

B. Swin-C2PSA

In remote sensing image object detection tasks, models are required not only to accurately identify small targets but also to capture complex spatial relationships within large-scale backgrounds. However, when processing large-scale high-resolution remote sensing imagery, the C2PSA module in the YOLO11n model suffers from high computational complexity and low efficiency. To address these limitations, this article designs a Swin-C2PSA module based on CSPNet and Swin

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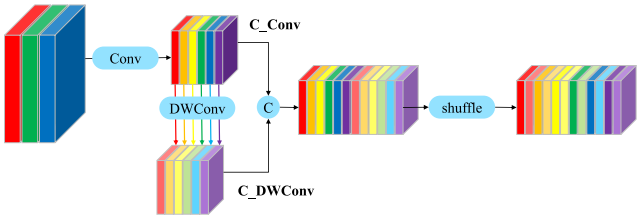


Fig. 3. GSConv module.

Transformer [16], [19], which replaces the original C2PSA module in YOLO11n.

The original C2PSA module employs a global self-attention mechanism, which requires computation over the entire feature map. Its computational complexity is $(4HWC^2 + 2(HW)^2C)$, where H, W, and C represent the height, width, and number of channels of the feature map, respectively. The larger the H and W of the feature map, the higher the complexity, and the lower the computational efficiency.

To address this issue, this article introduces the window-based multihead self-attention (W-MSA) and shifted window multihead self-attention (SW-MSA) mechanisms from Swin Transformer to design the Swin-C2PSA module. This module constrains attention within local windows, significantly reducing the self-attention computational complexity of the C2PSA module, thereby effectively improving the processing efficiency for large-scale remote sensing images. The Swin-C2PSA module divides the feature map into HW/M^2 windows, each of size $M \times M$. Its computational complexity is $\Omega(W - MSA) = 4HWC^2 + 2M^2HWC$ [16]. For large-scale remote sensing images, $HW \gg M^2$; therefore, the computational complexity of SW-MSA is approximately linear with respect to the image size $H \times W$ [16].

Given the computational efficiency advantages of the SW-MSA module, this article integrates it with the C2PSA module in YOLO11n to design the Swin-C2PSA module, whose structure is illustrated in Fig. 2. Following the design philosophy of CSPNet, the module first preprocesses the input feature map using a standard convolutional layer and then splits it along the channel dimension into two branches, namely, Split_A and Split_B. As shown in Fig. 2(b), Split_A is directly preserved through a cross-stage connection to retain shallow feature information, whereas Split_B is fed into the Swin-PSA module, which leverages shifted window-based attention to perform spatial feature extraction and cross-window interaction. Finally, Split_A and the processed Split_B are concatenated, and feature fusion is completed via a convolutional layer. The parallel structure adopted in this work serves two purposes. On the one hand, by restricting attention computation to local windows through SW-MSA, the module significantly reduces computational overhead when processing large-scale remote sensing imagery. On the other hand, shallow feature information is preserved through cross-stage connections, thereby achieving a balance between computational efficiency and feature integrity.

C. GS3k2

To balance recognition accuracy and computational efficiency in the task of road intersection detection over large-scale remote sensing images, this article introduces the GSConv module proposed by Slim-neck [18]. The structure of this module is illustrated in Fig. 3. Based on this module, this article designs an efficient feature aggregation module named GS3k2, whose structure is shown in Fig. 4. This module achieves lightweight model parameters while maintaining good feature representation capability.

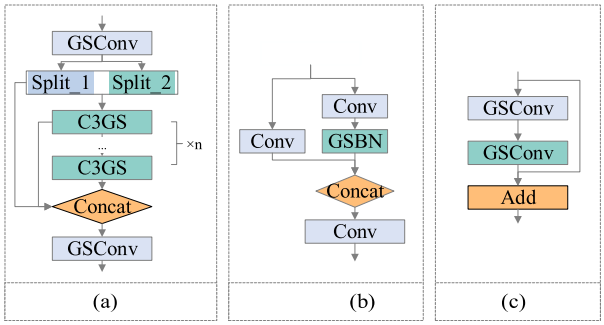


Fig. 4. Structure of GS3k2 module. (a) GS3k2. (b) C3GS. (c) GSBN.

As shown in Fig. 3, for a given input feature $X \in R^{C_1 \times H \times W}$ with C_1 channels, the GSConv module computes to produce the output feature Y with C_2 channels. The computational process is given as follows.

First, X passes through a standard convolution (Conv) to generate basic features (C_Conv) with $C_2/2$ channels; subsequently, DWConv, which has a lower computational cost, is used to generate redundant features (C_DWConv) with $C_2/2$ channels, and C_Conv and C_DWConv are concatenated to obtain the feature map D. Finally, a channel shuffle operation is applied to enable information interaction between features from different groups, ultimately yielding the feature map Y.

Y can be expressed as

$$Y = \text{Shuffle}\left(\text{Concat}\left[\text{Conv}_{SC}(X), \text{Conv}_{DC}(\text{Conv}_{SC}(X))\right]\right). \tag{1}$$

The computational complexity of the standard Conv is shown in the following equation:

$$\text{FLOPs}_{\text{sc}} = C_1 k^2 C_2 \times H'W'. \tag{2}$$

The computational complexity of GSConv is shown in the following equation:

$$\text{FLOPs}_{\text{GSConv}} \approx \left(C_1 \frac{C_2}{2} k_1^2 + \frac{C_2}{2} k_2^2\right) \times H'W' \tag{3}$$

where C_1 is the number of input feature channels, C_2 is the number of output feature channels, k is the size of the standard convolution filter, and k1 and k2 are the filter sizes in GSConv.

Based on the aforementioned GSConv module, this article further designs a lightweight GS3k2 module, whose structure is illustrated in Fig. 4(a). The workflow of the GS3k2 module is described as follows.

Step 1: GSConv is employed for preliminary feature extraction and channel adjustment, after which the feature map is split into two parts: Split_1 and Split_2.

Step 2: Split_1 is directly retained through a cross-stage connection to preserve shallow feature information, while

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Split_2 is fed into a stacked C3GS module [as shown in Fig. 4(b)] for deep spatial feature extraction.

Step 3: Split_1, the processed Split_2, and their intermediate features are concatenated, and final feature fusion is completed via GSConv.

The C3GS module mentioned in Step 2 is designed with two hierarchical components. First, a lightweight residual bottleneck structure, referred to as the GSBN module [shown in Fig. 4(c)], is constructed based on GSConv. Second, within the CSPNet framework, traditional convolutional layers and standard bottleneck structures are replaced with GSConv and GSBN modules, respectively, thereby forming a more lightweight C3GS module.

Furthermore, unlike the original approach that applies GSConv only in the Neck part [18], the proposed SW-GSE-YOLO model incorporates both GSConv and GS3k2 modules in the backbone and neck networks. This design not only significantly reduces the model's parameter count and computational burden but also effectively preserves feature representation accuracy.

D. Efficient-Intersection Over Union

Bounding box regression can adjust the prediction box to make it closer to the ground-truth box. The CIoU loss function [47] used in the YOLO11n model is shown in (4). Although this function comprehensively considers the overlap area, center point distance, and aspect ratio, it does not distinguish the true independent differences between width and height, leading to gradient direction conflicts during the optimization process, thereby slowing down convergence speed and affecting localization accuracy

$$L_{CIoU} = 1 - \mathrm{IoU} + \frac{\rho^2\left(b, b_{gt}\right)}{c_w^2 + c_h^2} + \frac{4}{\pi^2}\left(arctan\frac{w_{gt}}{h_{gt}} - arctan\frac{w}{h}\right)^2. \quad (4)$$

The EIoU loss function [20] is shown in (5). On the basis of retaining the advantages of CIoU, this function also considers directly regressing the differences in width and height between the target box and the prediction box, making the optimization direction clearer and effectively improving the model's convergence speed and detection localization accuracy

$$L_{EIoU} = 1 - \mathrm{IoU} + \frac{\rho^2\left(b, b_{gt}\right)}{c_w^2 + c_h^2} + \frac{\rho^2\left(w, w_{gt}\right)}{c_w^2} + \frac{\rho^2\left(h, h_{gt}\right)}{c_h^2}. \quad (5)$$

In (4) and (5), IoU represents the ratio of the overlapping area to the union area of the prediction box and the ground-truth box; b, w, and h represent the center point coordinates, width, and height of the prediction box, respectively. b_{gt}, w_{gt}, and h_{gt} represent the center point coordinates, width, and height of the ground-truth box, respectively. c_w and c_h represent the width and height of the minimum enclosing rectangle covering both the prediction box and the ground-truth box, respectively. $\rho(\cdot)$ represents the Euclidean distance. The schematic of each symbol is shown in Fig. 5.

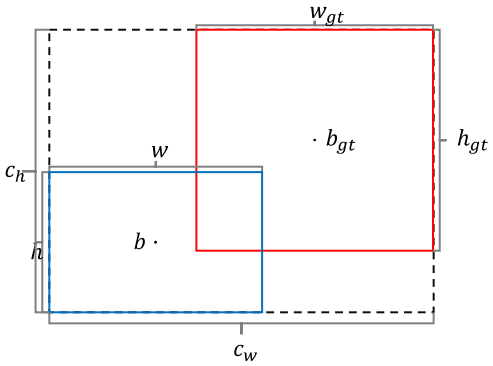


Fig. 5. Schematic of EIoU loss function.

IV. Experiments and Results

To comprehensively evaluate the performance of the proposed SW-GSE-YOLO model, this section first introduces the dataset, evaluation metrics, and experimental parameter settings. Subsequently, the effectiveness of each proposed improvement module is validated through ablation experiments, and performance comparisons are conducted against current mainstream object detection models. Finally, the generalization capability of the model across different geographical regions is demonstrated.

A. Setup

1) Dataset Details: To demonstrate the performance of the proposed SW-GSE-YOLO model in road intersection recognition, we constructed a dedicated road intersection dataset named RIS-42k. All imagery in this dataset is sourced from Google Earth, with a spatial resolution ranging from 0.5 to 1 m, ensuring sufficient clarity of ground object details. To cover diverse road scenarios, the dataset includes samples from different geographical regions, such as urban, suburban, and rural areas, and contains road intersections with various road network configurations.

During the annotation process, the center point of each road intersection was first determined, and a 400×400 pixel image patch was cropped with the center point as its center. To further expand the dataset, each image patch was rotated at 30° intervals, generating a total of 11 augmented samples per original patch and resulting in 42 372 images overall. All samples were divided into training, validation, and test sets with a ratio of 8:1:1. It should be noted that each original cropped sample and its corresponding rotated samples were assigned collectively to the same subset (training, validation, or testing) to ensure the reliability and fairness of the experiments. Fig. 6 presents several representative examples from the RIS-42k dataset.

2) Experimental Settings: The hardware system and software environment used in the experiment are given Table I.

We conducted experiments on the constructed road intersection dataset. The specific training parameter settings are given as follows: the number of epochs was set to 300, the optimizer selected was SGD with momentum, the momentum parameter was set to 0.9, and the initial learning rate was set to 0.01.

To improve training efficiency, the experiment was stopped if no performance improvement was observed after 15 epochs.

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Fig. 6. RIS-42k dataset intersection samples.

The first three epochs were used for warm-up, and the batch size was set to 64.

3) Metrics: To evaluate the performance of the SW-GSE-YOLO model in identifying and localizing road intersections in VHR imagery, this article employs evaluation metrics including precision, recall, mAP0.5, mAP at [0.5:0.95], model parameter count, and model size

$$IoU = \frac{Area\ of\ Overlap}{Area\ of\ Union} = \frac{A_{pred} \cap A_{gt}}{A_{pred} \cup A_{gt}} \tag{6}$$

$$Precision = \frac{TP}{TP + \mathrm{FP}} \tag{7}$$

$$Recall = \frac{TP}{TP + \mathrm{FN}}. \tag{8}$$

IoU is the degree of overlap between the predicted bounding box and the ground-truth bounding box. A detection is typically considered correct when IoU $\geq$ 0.5; true positive (TP): predictions with IoU $\geq$ threshold and correct classification; false positive (FP): predictions with IoU $<$ threshold or incorrect classification; and false negative (FN): ground-truth targets that were not detected. Precision represents the proportion of correct predictions among all predictions, which can measure the rate of FPs. Recall represents the proportion of correctly detected targets among all ground-truth targets, which can measure the rate of FNs.

The average of the average precision (AP) across all categories is

$$\mathrm{AP} = \int_0^1 \mathrm{Precision}\,(\mathrm{r})dr \tag{9}$$

$$mAP = \frac{1}{N}\sum_{i=1}^{N} AP_i. \tag{10}$$

1) *mAP0.5:* The mean AP calculated at an IoU threshold of 0.5.

TABLE I
EXPERIMENTAL ENVIRONMENT

Item	Configuration
Operating System	Ubuntu 20.04.6 LTS
GPU	Tesla T4 16G
Programming Language	3.10.11
Deep Learning Framework	Pytorch2.5.1、cuda11.8

TABLE II
COMPARISON EXPERIMENTS OF ATTENTION MECHANISMS

Attention	Precision	Recall	mAP0.5	mAP@[0.5:0.95]
ChannelAttention [37]	0.852	0.844	0.918	0.478
SpatialAttention [38]	0.836	0.842	0.912	0.477
CBAM [38]	0.836	0.826	0.898	0.438
CoordAttention [45]	0.797	0.761	0.846	0.415
Dual Attention [46]	0.834	0.811	0.895	0.448
C2PSA [17]	0.827	0.84	0.902	0.447
SwinC2PSA	**0.856**	**0.847**	**0.919**	**0.515**

2) *mAP at [0.5:0.95]:* The average mAP over IoU thresholds from 0.5 to 0.95 in steps of 0.05.

Since the focus of road intersection recognition lies in accurately identifying the location of their center points, mAP, as a comprehensive metric, can provide a more holistic evaluation of the model's performance in recognizing road intersections in VHR imagery. In addition, metrics such as model parameter count, model size, and computational complexity are used to evaluate model efficiency.

B. Results

To validate the effectiveness of each proposed module in the SW-GSE-YOLO model for road intersection detection in remote sensing imagery, we conducted comprehensive ablation experiments on the RIS-42k dataset under identical experimental conditions. Specifically, the performance contributions of the Swin-C2PSA attention mechanism, the GSConv and GS3k2 modules, and the IoU loss function were evaluated separately.

1) Swin-C2PSA Module: To verify the performance improvement brought by the Swin-C2PSA attention module, we compared it with several commonly used attention mechanisms, including channel attention from SENet [37], spatial attention and CBAM [38], coordinate attention [45], dual attention [46], and the original C2PSA module in the YOLO11n model. In these experiments, the C2PSA module in the YOLO11n baseline was replaced with each of the aforementioned attention mechanisms in turn, and all models were trained and tested under identical experimental settings. The corresponding results are reported in Table II.

The experimental results indicate that the original C2PSA module exhibits relatively limited performance in road intersection recognition tasks on remote sensing imagery, whereas the proposed Swin-C2PSA module achieves consistent improvements across all evaluation metrics. In particular,

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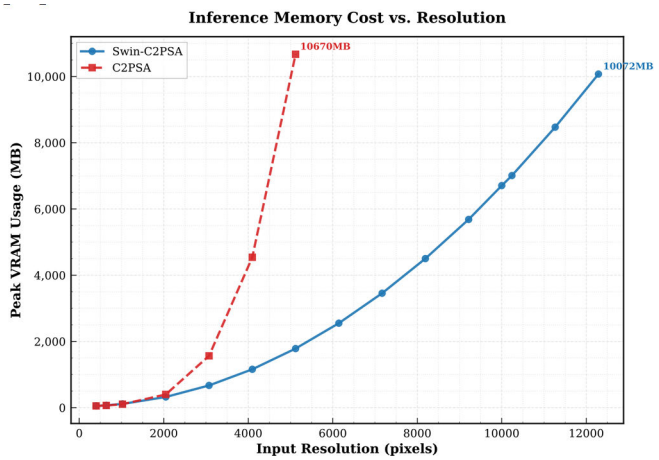


Fig. 7. Comparison of VRAM usage between the original C2PSA module in YOLO11n and the proposed Swin-C2PSA module.

TABLE III
EXPERIMENTS ON LIGHTWEIGHT MODULES

Modules	Precision	Recall	mAP 0.5	mAP@[0.5:0.95]	Parameters(M)	GFLOPs
Conv	0.827	0.84	0.902	0.447	2.6	6.5
DWConv [33]	0.824	0.83	0.898	0.499	1,9	4.8
GhostConv [35]	0.853	0.837	0.919	0.499	2.3	5.7
GSConv [18]	0.871	0.864	0.937	0.555	2.3	5.7
GSConv & GS3k2	**0.88**	**0.865**	**0.94**	**0.576**	**1.96**	**5.5**

TABLE IV
PERFORMANCE COMPARISON OF DIFFERENT IOU LOSS FUNCTIONS

IoU Type	Precision	Recall	mAP0.5	mAP@[0.5:0.95]
CIoU [47]	0.827	0.84	0.902	0.447
GIoU [48]	0.845	0.842	0.92	0.492
DIoU [47]	0.858	0.827	0.915	0.478
EIoU [20]	**0.871**	**0.853**	**0.935**	**0.538**

the mAP at [0.5:0.95] metric is improved by 6.8 percentage points, which effectively demonstrates the capability of the Swin-C2PSA module to enhance the model's road intersection feature extraction performance.

To further verify the advantages of the Swin-C2PSA module when processing large-scale remote sensing imagery, we compared the VRAM consumption of the original C2PSA module in YOLO11n with that of the proposed Swin-C2PSA module under different input image sizes. The corresponding experimental results are presented in Fig. 7. As shown in Fig. 7, the VRAM usage of the C2PSA module in the YOLO11n model increases rapidly as the input image size grows. When the input resolution reaches 6144×6144 pixels, an out-of-memory (OOM) error occurs on a Tesla T4 GPU with 16-GB memory.

In contrast, the proposed Swin-C2PSA module effectively reduces memory overhead by restricting self-attention computation to local windows, resulting in a much slower increase in VRAM usage as the input image size increases. With approximately 10 GB of available VRAM, the Swin-C2PSA module is still capable of successfully performing inference on remote sensing imagery with a resolution of up to $12\,000 \times 12\,000$ pixels. These results clearly demonstrate the superiority of the Swin-C2PSA module in handling high-resolution remote sensing imagery and significantly enhance the practical applicability of the proposed method.

TABLE V
ABLATION STUDY OF SW-GSE-YOLO

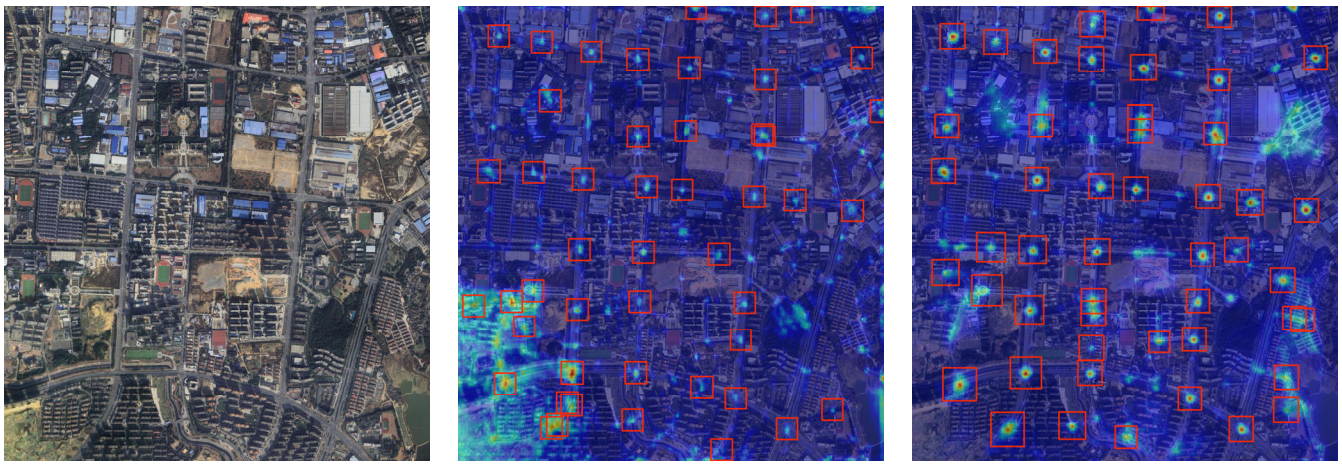
Swin-C2PSA	GSConv & GS3k2	EIoU	Precision	Recall	mAP0.5	mAP@[0.5:0.95]
			0.827	0.84	0.902	0.447
✓			0.856	0.845	0.919	0.515
	✓		0.88	0.865	0.94	0.576
		✓	0.871	0.853	0.935	0.538
✓	✓		0.844	0.815	0.906	0.533
✓		✓	0.859	0.85	0.925	0.565
	✓	✓	0.884	0.859	0.941	0.588
✓	✓	✓	**0.886**	**0.853**	**0.941**	**0.593**

2) GSConv and GS3k2 Module: To reduce computational complexity while maintaining model accuracy, this article introduces GSConv and its extended module GS3k2. We conducted comparative experiments between these modules and several commonly used convolutional variants, including standard convolution (Conv), depthwise separable convolution (DWConv) [33], GhostConv proposed by GhostNet [35], and GSConv [18]. The experimental results are reported in Table III. In these experiments, the corresponding convolutional modules were used to completely replace the original Conv modules in both the backbone and neck networks, while the GS3k2 module was employed to replace the original C3k2 module.

Although DWConv yields the lowest parameter count, its detection accuracy decreases accordingly. In contrast, the GSConv module introduces a channel shuffle mechanism on the basis of GhostConv. While the two modules exhibit comparable computational costs, GSConv achieves a substantial improvement in detection accuracy. Furthermore, after incorporating the GS3k2 module, the parameter count is reduced to 1.96 M, and the detection accuracy is further improved. These results demonstrate that the introduction of GSConv and GS3k2 modules can effectively reduce computational complexity and model size while preserving detection accuracy.

3) Different IoU Loss Functions: To evaluate the impact of different IoU loss functions on the accuracy of road intersection recognition and localization, this article compares the performance of CIoU [47], GIoU [48], DIoU [47], and EIoU [20] in the road intersection detection task based on the YOLO11n model. In this experiment, mAP at [0.5:0.95] is adopted as the primary evaluation metric, as it can more comprehensively reflect the overlap between predicted bounding boxes and ground-truth boxes under different IoU

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(a) (b) (c)

Fig. 8. Heatmap of YOLO11n and SW-GSE-YOLO. (a) Remote sensing image. (b) YOLO11n. (c) SW-GSE-YOLO.

TABLE VI
ACCURACY COMPARISON OF DIFFERENT MODELS FOR ROAD INTERSECTION EXTRACTION

Model	Precision	Recall	mAP0.5	mAP@[0.5:0.95]	Parameters(M)	Latency(*ms*)
yolov8n [17]	0.865	0.844	0.924	0.542	3.2	6.06
yolov8s [17]	0.878	0.848	0.932	0.548	28.6	5.21
yolov9t [49]	0.868	0.847	0.921	0.51	2	13.41
yolov10n [50]	0.852	0.837	0.919	0.49	2.3	8.17
yolo11n [17]	0.827	0.84	0.902	0.447	2.6	7.11
yolo11s [17]	0.858	**0.863**	0.93	0.549	9.4	7.09
yolov12n [51]	0.871	0.845	0.919	0.544	2.6	12.42
RT-DETR-l [52]	0.873	0.85	0.922	0.537	32.8	68.59
SW-GSE-YOLO(Ours)	**0.886**	**0.853**	**0.941**	**0.593**	**1.8**	**10.04**

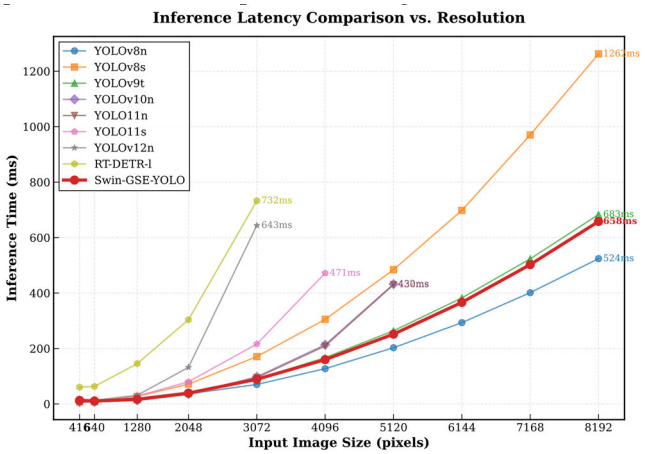


Fig. 9. Model inference latency versus image size.

thresholds. The corresponding experimental results are reported in Table IV.

The experimental results indicate that EIoU outperforms all other IoU loss functions across all evaluation metrics in the road intersection detection task. In particular, the mAP at [0.5:0.95] metric increases by 9.1% compared with CIoU, corresponding to a relative improvement of 20.3%. These results demonstrate that EIoU can effectively enhance the localization accuracy of road intersections.

TABLE VII
RECOGNITION PRECISION AND RECALL FOR DIFFERENT CITIES AND REGIONS

Region	TP	FP	FN	Precision	Recall
BeiJing	146	9	11	0.942	0.93
NanChang	176	9	25	0.951	0.876
Suburban	53	4	6	0.93	0.898
Rural	28	2	5	0.933	0.848

C. Ablation Study

To validate the synergistic effects of the various improvement strategies in the proposed SW-GSE-YOLO model, this article conducts a series of ablation experiments using YOLO11n as the baseline model, in which the GSConv/GS3k2 module, the Swin-C2PSA module, and the EIoU loss are

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Fig. 10. Recognition results of road intersections in VHR remote sensing images in different regions. (a) Beijing (main urban area). (b) Nanchang (main urban area). (c) Suburban area. (d) Rural area.

sequentially incorporated. The corresponding results are reported in Table V.

The experimental results indicate that the introduction of the GSConv and GS3k2 modules yields the most substantial performance improvement among all strategies. Although certain evaluation metrics exhibit fluctuations when GSConv/GS3k2 and Swin-C2PSA are introduced simultaneously, which may be attributed to the difficulty of the lightweight backbone network in rapidly adapting to the high-level semantic features extracted by the Swin-C2PSA module, the overall performance is restored and further enhanced with the additional incorporation of the EIoU loss. Ultimately, the proposed SW-GSE-YOLO model, which integrates all improvement strategies, consistently outperforms the original YOLO11n baseline across all evaluation metrics.

D. Comparisons and Analysis

To qualitatively evaluate the effectiveness of the proposed SW-GSE-YOLO model, the Grad-CAM method [49] is employed to visualize the activation heatmaps of the baseline YOLO11n model and the proposed SW-GSE-YOLO model. The corresponding results are presented in Fig. 8. As shown in the figure, SW-GSE-YOLO exhibits stronger and more focused responses on road intersection regions compared with the YOLO11n model, indicating its superior perception capability for road intersections.

To comprehensively evaluate the performance of the proposed SW-GSE-YOLO model, this article compares it with several mainstream YOLO-based object detection models and the real-time DEtection TRansformer (RT-DETR-l) model on

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the RIS-42k dataset. The corresponding results are reported in Table VI.

As shown in Table VI, the proposed SW-GSE-YOLO model has the smallest parameter count among all compared methods, with only 1.8M parameters. Compared with YOLOv8n, YOLOv8s, YOLOv9t, YOLOv10n, YOLO11n, YOLO11s, YOLOv12n, and RT-DETR-l, the parameter count of SW-GSE-YOLO is reduced by 43.75%, 93.71%, 10.00%, 21.74%, 30.77%, 80.85%, 30.77%, and 94.51%, respectively. Despite this substantial reduction in model parameters, SW-GSE-YOLO achieves a higher precision than all compared models in the road intersection detection task, while its recall is only slightly lower than that of the larger YOLO11s model. Moreover, with respect to the two key evaluation metrics, mAP at 0.5 and mAP at [0.5:0.95], SW-GSE-YOLO outperforms all comparison methods, demonstrating an excellent balance between detection accuracy and model complexity.

In terms of inference efficiency, when evaluated on the RIS-42k dataset with an input size of 400 × 400 pixels, the SW-GSE-YOLO model achieves an inference latency of 10.04 ms. Its inference speed is faster than that of YOLOv9t, YOLOv12n, and RT-DETR-l while being slightly slower than the other YOLO-series models included in the comparison. However, since the inference latency reported in Table VI is measured based on the 400 × 400 pixel RIS-42k dataset, it is difficult to fully reflect the inference performance of the SW-GSE-YOLO model when processing large-scale high-resolution remote sensing imagery.

Therefore, under identical experimental conditions, remote sensing images of different sizes were selected to further evaluate the inference latency performance of the SW-GSE-YOLO model and the comparison models. The corresponding experimental results are presented in Fig. 9.

As shown in Fig. 9, the inference latency of different models exhibits a clear positive correlation with the size of the input remote sensing imagery. When the input resolution ranges from 416 × 416 to 3072 × 3072 pixels, all compared models are able to achieve relatively fast inference speeds. However, as the image size further increases beyond 3072 × 3072 pixels, some models begin to encounter OOM issues. For instance, when processing remote sensing imagery with resolutions larger than 4096 × 4096 pixels, RT-DETR-l, YOLO11s, and YOLOv12n fail to complete the inference task due to OOM errors. When the input resolution reaches 8192 × 8192 pixels, only YOLOv8n, YOLOv8s, YOLOv9t, and the proposed SW-GSE-YOLO model are able to run stably. Among these models, although the inference latency of SW-GSE-YOLO is slightly higher than that of YOLOv8n, it demonstrates a stronger capability for processing large-scale remote sensing imagery while maintaining a lower parameter count and higher detection accuracy. This further verifies the advantages of the proposed method in practical application scenarios.

To verify the generalization capability of the proposed SW-GSE-YOLO model across different regional scenarios, this article selected representative areas from the main urban regions of Beijing and Nanchang, as well as suburban and rural regions of Shangrao City, as test areas. The image resolution for the main urban areas of Beijing and Nanchang is 7644 × 7644 pixels, while that for the suburban and rural areas of Shangrao City is 4096 × 4096 pixels. All images used in this experiment are very high-resolution (VHR) remote sensing imagery, and representative examples are illustrated in Fig. 10(a)–(d). The corresponding experimental results are reported in Table VII.

As shown in Table VII, the proposed SW-GSE-YOLO model maintains stable and strong detection performance across different regional and scenario conditions. In the large-scale image scenarios of the main urban areas of Beijing and Nanchang, the model achieves precision values of 0.942 and 0.951, with corresponding recall values of 0.930 and 0.876, respectively. In the relatively smaller image scenarios of suburban and rural areas, the precision values are 0.930 and 0.933, while the recall values reach 0.898 and 0.848, respectively.

These experimental results indicate that the proposed SW-GSE-YOLO model exhibits robust detection stability and strong generalization capability when performing road intersection recognition and extraction tasks on VHR remote sensing imagery of varying image sizes and across different geographical regions. Consequently, the model can effectively adapt to road intersection detection requirements in diverse remote sensing scenarios.

V. Conclusion

This article presents SW-GSE-YOLO, a lightweight yet accurate detection model specifically designed for road intersection recognition in high-resolution remote sensing imagery. The proposed model effectively addresses several key challenges, including road intersection detection in large-scale remote sensing imagery, limited computational resources, and insufficient localization accuracy of road intersection center points.

Specifically, SW-GSE-YOLO is constructed through innovations at three levels. First, by incorporating the Swin Transformer architecture, a Swin-C2PSA module is designed to enhance the model's perception of local details and contextual information related to road intersections while enabling efficient modeling of high-resolution feature maps via window-based self-attention. Second, GSConv is introduced and combined with the CSP structure to design the C3GS and GS3k2 modules, thereby constructing an efficient lightweight network architecture that significantly reduces model parameters and computational complexity while preserving strong feature representation capability. Third, the adoption of the EIoU loss function optimizes the bounding box regression process, effectively improving the localization accuracy of road intersection center points.

Extensive experimental results on the self-constructed RIS-42k dataset fully validate the effectiveness of the proposed method. The results demonstrate that SW-GSE-YOLO provides an efficient and practical solution for road intersection detection in VHR remote sensing imagery. In particular, in large-scale remote sensing image scenarios, the proposed model significantly reduces VRAM consumption and inference latency.

Despite its strong performance across various scenarios, the proposed model still has certain limitations. When road intersections are heavily occluded by vegetation or buildings,

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missed detections may still occur. In addition, complex transportation hubs (e.g., interchanges) that exhibit high structural similarity to road intersections can lead to recognition ambiguities. Furthermore, due to variations in data distribution and diverse road design styles, there remains room for improvement in generalizing the model to global road networks and complex geographical environments. These limitations point to several promising directions for future research.

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This article does not contain any studies with human participants performed by any of the authors.

CONFLICT OF INTERESTS

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The datasets generated and/or analyzed during this study are available from the corresponding author on reasonable request, or can be accessed via the following link: https://github.com/tyuhao/RIS-42k.

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